Research Notes on Perplexity language model

Perplexity language model

>> Section 1

where x ranges over the events, where 0–0 is defined to be 1, and where the value of b does not affect the result; b can be chosen to be 2, 10, e, or any other positive value other than 1. In some contexts, this measure is also referred to as the (order-1 true) diversity. The logarithm log PP(p) is the entropy of the distribution; it is given the unit shannon (bit) if the base of the logarithm is 2, and nat if the natural logarithm is used. Perplexity of a random variable X may be defined as the perplexity of the distribution over its possible values x. It can be thought of as a measure of uncertainty or "surprise" related to the outcomes. For a probability distribution p where exactly k outcomes each have a probability of 1 / k and all other outcomes have a probability of zero, the perplexity of this distribution is simply k. This is because the distribution models a fair k-sided die, with each of the k outcomes being equally likely. In this context, the perplexity k indicates that there is as much uncertainty as there would be when rolling a fair k-sided die. Even if a random variable has more than k possible outcomes, the perplexity will still be k if the distribution is uniform over k outcomes and zero for the rest. Thus, a random variable with a perplexity of k can be described as being "k-ways perplexed", meaning it has the same level of uncertainty as a fair k-sided die. Perplexity is sometimes used as a measure of the difficulty of a prediction problem. It is, however, generally not a straightforward representation of the relevant probability. For example, if you have two choices, one with probability 0.9, your probability of a correct guess using the optimal strategy is 0.9. Yet, the perplexity is 0.9–0.9 0.1–0.1 = 1.38. The inverse of the perplexity, 1/1.38 = 0.72, is not equal to the probability 0.9. The perplexity is the exponentiation of the entropy, a more commonly encountered quantity. Entropy measures the expected or "average" number of bits required to encode the outcome of the random variable using an optimal variable-length code. It can also be regarded as the expected information gain from learning the outcome of the random variable, providing insight into the uncertainty and complexity of the underlying probability distribution.

>> Section 2

H (p ~ , q) = – ∑ x p ~ (x) log b ■ q (x) , {\displaystyle H({\tilde {p}},q)=-\sum _{x}{\tilde {p}}(x)\log _{b}q(x),}

>> Section 3

Perplexity of a probability model A model of an unknown probability distribution p may be proposed based on a training sample that was drawn from p. Given a proposed probability model q, one may evaluate q by asking how well it predicts a separate test sample x1, x2, ..., xN also drawn from p. The perplexity of the model q is defined as

>> Section 4

b – 1 N ∑ i = 1 N log b ■ q (x i) = (∏ i q (x i)) – 1 / N , {\displaystyle b^{-{\frac {1}{N}}\sum _{i=1}^{N}\log _{b}q(x_{i})}=\left(\prod _{i}q(x_{i})\right)^{-1/N},}

>> Section 5

Brown Corpus The lowest perplexity that had been published on the Brown Corpus (1 million words of American English of varying topics and genres) as of 1992 is indeed about 247 per word/token, corresponding to a cross-entropy of log2 247 = 7.95 bits per word or 1.75 bits per letter using a trigram model. While this figure represented the state of the art (SOTA) at the time, advancements in techniques such as deep learning have led to significant improvements in perplexity on other benchmarks, such as the One Billion Word Benchmark. In the context of the Brown Corpus, simply guessing that the next word is "the" will achieve an accuracy of 7%, contrasting with the 1/247 = 0.4% that might be expected from a naive use of perplexity. This difference underscores the importance of the statistical model used and the nuanced nature of perplexity as a measure of predictiveness. The guess is based on unigram statistics, not on the trigram statistics that yielded the perplexity of 247, and utilizing trigram statistics would further refine the prediction.

>> Section 6

Perplexity of a probability distribution The perplexity PP of a discrete probability distribution p is a concept widely used in information theory, machine learning, and statistical modeling. It is defined as

>> Section 7

The second metric, perplexity (of a token), is an information theoretic measure that evaluates the similarity of proposed model m to the original distribution p. It can be computed as a inverse of (geometric) average probability of test set T

>> Section 8

P P L (D) = 1 m (T) N = 2 – 1 N log 2 ■ (m (T)) , {\displaystyle \mathrm {PPL} (D)={\sqrt[{N}]{\frac {1}{m(T)}}}=2^{-{\frac {1}{N}}\log _{2}\left(m(T)\right)},}

>> Section 9

P P (p) = ∏ x p (x) – p (x) = b – ∑ x p (x) log b ■ p (x) , {\displaystyle \mathrm {PP} (p)=\prod _{x}p(x)^{-p(x)}=b^{-\sum _{x}p(x)\log _{b}p(x)},}

>> Section 10

where b {\displaystyle b} is customarily 2. Better models q of the unknown distribution p will tend to assign higher probabilities q(xi) to the test events. Thus, they have lower perplexity because they are less surprised by the test sample. This is equivalent to saying that better models have higher likelihoods for the test data, which leads to a lower perplexity value. The exponent, – 1 N ∑ i = 1 N log b ■ q (x i) {\textstyle -{\frac {1}{N}}\sum _{i=1}^{N}\log _{b}q(x_{i})} , with b = 2, may be regarded as the average number of bits needed to represent a test event xi if one uses an optimal code based on q. Low-perplexity models do a better job of compressing the test sample, requiring few bits per test element on average because q(xi) tends to be high. The exponent may also be interpreted as a cross-entropy: